Modeling and Forecasting Black Friday Search Volume Index Using Time Series Regression and Hybrid ARIMA Models

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Abstract. Black Friday is an annual event that causes a significant surge in online search activity. This phenomenon causes Search Volume Index (SVI) data to have a very strong seasonal pattern, extreme calendar variation, and non-stationary characteristics. This study aims to model and predict Black Friday SVI using a time series approach that combines deterministic and stochastic components. The methods used include Time Series Regression (TSR), regression with ARIMA errors (ARIMAX), and a hybrid TSR model with ARMA errors. Several model specifications were compared based on model structure, prediction accuracy in training and testing data, and residual diagnostic test results. The analysis results show that models with seasonal components and calendar variations are better at capturing annual surge patterns than simple models. In addition, the hybrid model that combines deterministic regression and ARMA errors provides more stable prediction performance and meets the white noise residual assumption. Based on a comprehensive evaluation, the hybrid model was selected as the best model for Black Friday SVI forecasting. The forecasting results show that search spikes are expected to continue to occur consistently during Black Friday periods in the future.

Keywords: Black Friday · Search Volume Index · Calendar Variation · ARIMAX · Time Series Regression

1 Introduction

Black Friday is the day after Thanksgiving. It is the biggest shopping day of the year in the United States. Both brick-and-mortar and online retailers offer special deals, promotions, and discounts on many products. One theory suggests that the day is called Black Friday because it is when many retailers begin to turn a profit (move into the black) for the year [1]. This idea is borrowed from the casual use of the terms “in the red” to describe, in general, a state of loss, and “in the black” to refer to making a profit [2].

With the development of technology and online commerce or e-commerce, the concept of Black Friday has transformed into a global phenomenon involving various countries, including Indonesia. Black Friday is not only a culture for Americans, but has become a global culture that has spread to various countries, including Indonesia [3]. In Indonesia, there are discounts of up to 80% during Black Friday on platforms such as Zalora, Expedia, Agoda, ASOS, Samsung, and Sephora [4]. The most frequently searched e-commerce site is Lazada at 32.25% [5].

In addition, searches for “Black Friday” also reveal interesting geographical patterns in Indonesia. Data shows that the regions with the highest searches for the term Black Friday are Bali, followed by Riau, DKI Jakarta, and the Special Region of Yogyakarta (DIY) (Google Trends, 2025). This decision to shop online is influenced by convenience, trust, and lifestyle [6]. This can be seen in Bali, which is an international tourist destination and a center of global economic activity, so its residents are more accustomed to global shopping trends. In addition, Yogyakarta, which is a student city, is also proof that 30% of Indonesians aged 18-25 shop online, contributing a significant percentage [7]. This distribution pattern indicates that the Black Friday phenomenon in Indonesia is not only temporary and seasonal but is also influenced by geographical, economic, and demographic characteristics of each region.

Based on the temporary increase in Black Friday searches in Indonesia, which is influenced by calendar variations and shows regional differences, there is a need to dig deeper into the search patterns of the public. However, it is still unclear how the trends and seasonality in Black Friday search data in Indonesia are structured, whether they are better modeled as deterministic trends or seasonality, or perhaps stochastic trends or seasonality. In addition, the effect of variations in the Black Friday calendar, which is an annual event, on search volume also needs to be tested using an empirical approach. Differences in modeling methods, from ordinary trend models, models with seasonal components, to models involving ARMA elements in errors, raise questions about which model is most effective in accurately representing and predicting Black Friday search patterns in Indonesia based on data from Google Trends.

Therefore, this study aims to explore and compare various time series models in projecting the number of Black Friday keyword searches in Indonesia using data from Google Trends. The focus of the analysis is on identifying trend and seasonal patterns using deterministic and stochastic approaches, including seasonal Fourier analysis and the influence of Black Friday calendar variations. In addition, this study also aims to identify the most effective model that can represent people's search behavior, so that it can be used as a reference for economic actors and e-commerce platforms in formulating more appropriate and efficient promotional strategies.

2 Literature Review

2.1 Time Series

Time series is defined as a sequence of observations recorded in chronological order, where each observation represents the value of a variable at a specific point in time. Time series data are generated by complex underlying processes and therefore exhibit structured temporal patterns rather than purely random behavior [8]. These patterns reflect the dynamic evolution of the observed phenomenon and must be analyzed by considering the dependency between observations over time.

2.2 Trend

Trend is defined as a systematic long-term movement in the level of a series that causes its statistical properties, particularly the mean, to change over time. The presence of trend implies non-stationarity, meaning that the probabilistic behavior of the series is not invariant with respect to time. Two conceptually different forms of trend can be distinguished: deterministic trend and stochastic trend. A deterministic trend refers to a systematic, predictable evolution of the mean over time, often represented conceptually as a fixed function of time plus a stationary disturbance. In contrast, a stochastic trend arises when the non-stationarity is generated by the accumulation of random shocks over time.

2.3 Seasonal

Seasonality in time series refers to systematic and recurrent patterns that occur at regular intervals, such as monthly or yearly cycles. Seasonal behavior is a fundamental characteristic of many real-world time series and must be carefully modeled to obtain accurate interpretation and forecasting results [9]. Seasonal patterns can be represented either as deterministic structures or as stochastic processes, depending on the stability and complexity of the seasonal behavior observed in the data. Deterministic seasonality assumes that seasonal patterns are fixed and repeat identically over time. In contrast, stochastic seasonality treats the seasonal component as a dynamic process that evolves gradually through time. Deterministic seasonality is suitable for simple and stable seasonal patterns, whereas stochastic seasonality offers greater flexibility and robustness for complex, evolving, and irregular seasonal behavior.

2.4 Fourier Series

Fourier series provide a flexible and parsimonious way to represent seasonal patterns in time series by decomposing periodic behavior into a sum of sinusoidal components. Seasonality can be interpreted as a deterministic periodic signal that may be expressed through trigonometric functions with known frequencies [10]. In this framework, seasonal variation is approximated by a combination of sine and cosine terms, each associated with a specific harmonic of the fundamental seasonal period.

$$s(t) = \sum_{n=1}^{N} \left(a_n cos\left(\frac{2\pi nt}{p}\right) + b_n sin\left(\frac{2\pi nt}{p}\right) \right) \quad (1)$$

where P is time periodic.

2.5 Calendar Variation

Calendar variation refers to systematic fluctuations in time series data that arise from irregularities in the calendar structure rather than from inherent economic or stochastic processes. Calendar variation occurs because calendar units such as months and quarters are not uniform in length and because certain events recur according to the calendar rather than at fixed time intervals [11]. As a result, observed time series values may vary depending on the number of working days, weekends, or specific holidays contained within a given period. Proper adjustment for calendar effects improves the accuracy of decomposition, enhances model interpretability, and leads to more reliable forecasting results by preventing calendar-related distortions from being misinterpreted as genuine seasonal or trend movements.

2.6 Time Series Regression

Time series regression is a modeling approach used to describe the relationship between a dependent variable observed over time and one or more explanatory variables that may also vary across time. Time series regression is employed to model temporal data that exhibit systematic components such as trend, seasonality, and calendar-related effects. This approach explicitly incorporates deterministic structures through regression terms, allowing external or structural factors to be directly represented within the model. Time series regression can accommodate a trend component by including time as an explanatory variable, typically represented as a linear function of the observation index. This formulation allows long-term movements in the data to be captured through a deterministic trend term, while the remaining unexplained variation is attributed to the error component

to satisfy classical regression assumptions, particularly independence, constant variance, and normality. Seasonal dummy variables represents a specific seasonal period, such as a month, enabling the model to capture systematic recurring fluctuations [11]. This approach allows seasonal effects to be modeled explicitly rather than inferred indirectly from autoregressive structures, making the interpretation of seasonal influences more transparent.

2.7 ARIMA

Autoregressive Integrated Moving Average (ARIMA) (p,d,q) model is a combination of the AR and MA models after differencing at the *d* order. ARIMA is a fairly popular method for conducting time series analysis. One of the requirements that must be met in forming an ARIMA model (based on the AR and MA orders) is that the data must be stationary in both means and variance. ARIMA modeling can be done using the Box-Jenkins method, with the following sequence: model identification, parameter estimation, assumption testing, and forecasting. In general, the ARIMA model can be written as follows [12].

$$\phi_p(B)(1-B)^d Y_t = \theta_0 + \theta_9(B)a_t \tag{2}$$

Where

$$\phi_p(B) = \left(1 - \phi_1 B - \cdots - \phi_p B^p\right) \tag{3}$$

$$\theta_q(B) = \left(1 - \theta_1 B - \cdots - \theta_q B^q\right) \tag{4}$$

2.8 ARIMAX

Autoregressive Integrated Moving Average with Exogenous variables (ARIMAX) model is an extension of the classical ARIMA framework that incorporates additional external or explanatory variables into the time series modeling process. ARIMAX was developed to overcome the limitations of univariate ARIMA models by allowing the inclusion of exogenous factors that influence the behavior of the dependent time series [13]. In ARIMAX, the target time series is modeled not only as a function of its own past values and past forecast errors but also as a function of one or more external variables that are assumed to have explanatory power. These exogenous variables may represent parallel time series, such as economic indicators, meteorological variables, or policy-related factors, which are correlated with the response variable and help improve forecasting accuracy.

2.9 Residual Diagnostic Test

Residual diagnostic testing constitutes a crucial step in validating the adequacy of a fitted time series model. After a model has been identified, estimated, and fitted to the data, the remaining residuals are expected to contain no systematic information. Model is considered adequate only if the residual series behaves like white noise, indicating that the model has successfully captured all meaningful structure present in the original data [14]. White noise residuals are defined as a sequence of random variables with a zero mean, constant variance, and no serial correlation across time. The presence of white noise residuals implies that the fitted ARIMA model has effectively removed all autocorrelation and systematic patterns from the time series.

2.10 Metric Evaluation

Evaluating the accuracy of point forecasts is essential to assess how well a model predicts future observations. Forecast errors are defined as the difference between observed values and the corresponding forecasts. These errors are then summarized using various accuracy metrics that quantify how far the forecasts deviate from the actual values.

RMSE

Root Mean Squared Error (RMSE) is a scale-dependent measure that quantifies the square root of the average of squared forecast errors. It is given by the square root of the mean of the squared differences between the forecasted and actual values. Lower values of RMSE indicate more accurate forecasts, with a perfect forecast yielding an RMSE of zero.

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{5}$$

MAPE

Mean Absolute Percentage Error (MAPE) represents the average of the absolute forecast errors expressed as a percentage of the observed values, used to measure the average deviation of the forecast from the actual data, expressed as a percentage. A small MAPE value indicates that the forecasted data is close to the actual value.

$$MAPE = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{6}$$

sMAPE

Symmetric Mean Absolute Percentage Error (sMAPE) was introduced to address some of the limitations of MAPE, particularly its asymmetry and instability when actual values are small. sMAPE normalizes the absolute forecast error by the average of the absolute actual and forecasted values, thereby producing a bounded and more symmetric error measure, with smaller values indicating better forecasting performance.

$$\text{sMAPE} = \frac{100}{n}\sum_{t=1}^{n}\frac{|y_t - \hat{y}_t|}{(|y_t| + |\hat{y}_t|)/2} \tag{7}$$

2.11 Forecasting

Forecasting in time series analysis refers to the process of predicting future observations based on historical data and the underlying structure of the series. Forecasts are generated by extrapolating some components like trend, seasonal effects, and irregular disturbances in time using estimated model parameters. This approach allows the forecast distribution to account not only for the expected future value but also for the uncertainty associated with each component of the model.

3 Analysis and Discussion

3.1 Data Description and Dataset Characteristics

This study uses a dataset from Google Trends. The dataset is a Search Volume Index (SVI) for the keyword "Black Friday" in Indonesia. The data is arranged in a monthly time series with a period from January 2015 to December 2025, with a total of 132 observations. Google Trends presents data on a relative scale of 0 – 100, where a value of 100 indicates the highest level of search interest in the observed time period, not the actual number of searches [15]. The main characteristics of the data show a very strong event–based seasonal pattern, with a sharp spike every November, as well as a long-term trend that tends to increase gradually [8].

3.2 Variable Identification and Model Specification

Research variables were identified to capture trend, seasonal, and calendar variations.

Table 1. Research Variables

Category	Variables	Symbol	Description
Data Variable	SVI Black Friday	y_t	Main variable being forecasted.
New Variable	Time (Trend)	t	Time sequence index.
	Month (Dummy)	D_{it}	Binary variable (1 or 0) for each month.
	Fourier Terms	s_k, c_k	Sine/cosine functions for smooth seasonal patterns.
	Calendar Effect	C_t	Binary variable (1 or 0) for spikes in November.
Model Components	Residual	η_t	Model Residuals contain autocorrelation patterns.

Model specification:

This study tested eight models, which can be divided into three groups as follows.

1) Group 1 : Classic Time Series Regression

- Trend Only
  The basic model that only look at trend patterns, whether there is a linear increase or decrease in search interest from 2015 to 2025 [16].
- Trend + Seasonal
  This model adds a dummy variable to capture the effect of each month or seasonal (January-December) [16].
- Trend + Seasonal + Calendar (Black Friday)
  This model builds on the previous model by adding a calendar to variation capture the extreme spikes in November [17].

2) Group 2 : Dynamic Regression

- Deterministic Trend and Deterministic Seasonality (Fourier)
  This model uses sine and cosine functions because it uses Fourier to replace the dummy variable so that the seasonal pattern appears smoother [16].
- Stochastic Trend and Deterministic Seasonality (Fourier)
  This model uses sine and cosine functions, and the trend is not considered a straight line. The model allows the trend to change through a *differencing* process [16].
- Deterministic Trend and Stochastic Seasonality
  This model uses a fixed linear trend but allows seasonal patterns to change each year using the ARIMA (SARIMA) seasonal component [16].

- Stochastic Trend and Seasonality
  This model does not assume a straight line for the trend or a fixed pattern for seasonality, so it follows the latest data movement [14].

3) Group 3 : Hybrid Models

- Hybrid Time Series Regression + ARMA Error
  This model is a two-stage combination. The first stage explains the data through Trend + Seasonal + Calendar (Black Friday), while the second stage corrects prediction errors using past residual patterns [16].

3.3 Model Evaluation

This stage is necessary to test the extent to which the model can capture past data patterns to predict future data. This aims to ensure the model's generalization ability and avoid overfitting, where the model is only accurate on training data but fails when making actual predictions.

Train-Test Split

Data division in time series cannot be done randomly. This is because there is temporal dependence between observations whose order must be maintained [16]. Therefore, the dataset is divided chronologically into two main parts.

1) Training Data
   The training data forms the basis for the model to recognize historical trends and seasonal patterns. This data covers the period from January 2015 to December 2024. The training data is used to estimate model parameters, determine the optimal ARMA order and calculate the regression coefficients for the eight models.
2) Test Data
   The test data covers the period from January 2025 to December 2025. This data is used as comparative data. The model makes forecasts for the 2025 period without "seeing" the actual data. The forecast results are then compared with the 2025 data to calculate the error rate or model accuracy using the RMSE, MAPE and sMAPE metrics.

Accuracy Evaluation

This study uses RMSE, MAPE and sMAPE metrics to compare the performance between models.

1) Root Mean Square Error (RMSE)
   RMSE calculates the root of the means square difference between actual (y_t) and predicted values ($\hat{y}_t$).

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{8}$$

   RMSE is effective for detecting how accurately the model captures the peak that occurs in November [14].
2) Mean Absolute Percentage Error (MAPE)
   MAPE measures the average absolute percentage error relative to the actual value.

$$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{9}$$

MAPE has a relative scale (percentage), making the quality of the forecast easier to understand [16].

3) Symmetric Mean Absolute Percentage Error (sMAPE)
sMAPE is to address the limitations of MAPE, sMAPE is a symmetric metric that uses the average of the actual and predicted value in the divisor.

$$\text{sMAPE} = \frac{100\%}{n}\sum_{t=1}^{n}\frac{|y_t - \hat{y}_t|}{(|y_t| + |\hat{y}_t|)/2} \tag{10}$$

sMAPE provides a more balanced accuracy measure, especially when the dataset contains values close to zero. This metric prevents excessive penalty bias in prediction results that are higher than the actual values.

3.4 Diagnostic Model Procedure

After the best model has been selected based on the criteria of minimum accuracy of RMSE and MAPE in the test data, the next step is to perform a residual diagnostic test. The purpose is to ensure that the model has captured all systematic patterns in the data (trends, seasonality, and calendar), so that the residual error is random and meets the white noise assumption [14]. Residual that meet this assumption indicate that the model is optimal and no significant information has been left out.

Visual Analysis (ACF Plots)

This initial diagnostic stage is performed visually by examining the Autocorrelation Function (ACF) plot of the residuals. This plot is used to detect correlation between time series in the residuals at various lags. The assumption of white noise is satisfied if all spikes on the ACF plot are within the significance limits [16]. If there are spikes outside the significance limits, this indicates that there are still residual patterns that can be modeled further.

Ljung -Box Test

This test evaluates whether the residual autocorrelation in a set of lags is simultaneously different from zero [14].

Test Hypothesis:

$H_0: \rho_1 = \rho_2 = \cdots = \rho_h = 0$ (residuals are white noise)

$H_1: at\ least\ one\ \rho_k \neq 0$ (residuals contain autocorrelation)

Test Statistics:

$$Q = n(n+2)\sum_{k=1}^{h}\frac{\hat{\rho}_k{}^2}{n-k} \tag{11}$$

Where n is the sample size, $\hat{\rho}_k{}^2$ is the autocorrelation at lag k and h is the numbers of lags tested.

Decision Criteria:

If p-value > 0.05, then fail to reject, which means the residuals are white noise and the model has captured all systematic patterns [14].

Model Validation

The model passes the diagnostic test if it meets the assumption ACF and Ljung-Box. Success in this stage ensures that:

1) The model optimally represents the characteristics of the data (trends, seasonality, and calendar).
2) Parameter estimates are efficient and unbiased.

3) The model has high reliability for use in forecasting future periods (January 2026-December 2026).

3.5 Forecasting Framework

After the best model was selected through a series of evaluation and diagnostic tests, the final stage of this research was forecasting. This stage aimed to provide quantitative estimates of future Black Friday search trends.

Criteria for Selecting the Best Model

1) Lowest Accuracy
   The model is selected based on the smallest RMSE, MAPE, and sMAPE values in the test data (out-of-sample), in accordance with the evaluation in the predicted test object [16].
2) Residual Validity
   The model must produce white noise residuals, verified through the Ljung-Box test with p -value > 0.05 [14].
3) Parsimony Principle
   If two models show nearly identical accuracy performance, the model with a simpler parameter structure (e.g., pure linear regression versus hybrid models) is selected to maintain the stability and interpretability of the forecasting results [16].

Horizon and Future Data Preparation

Forecasting is performed for a short-term horizon of 12 months, namely the period from January 2026 to December 2026. A new dataset is created to include all the necessary predictor variables:

- Trend (t): continuation of the historical time series,
- Seasonal Variation: dummy variables for each month (January–December)
- Calendar Variation: The Black Friday variable is given a value of 1 in November 2026 and 0 for all other months.

Hybrid Forecasting Procedure

For the best hybrid model, forecasting is performed through two main components:

1) Deterministic Component
   Generates predictions based on a regression model which captures the effects of trends, seasonality, and Black Friday variables.
2) Stochastic Component
   Estimates residuals using an ARIMA model to capture temporal dynamics not captured by the regression component.
   The two components are combined to form a comprehensive final point forecast, covering all structural and stochastic information in the data.

Prediction Interval Estimates

Forecasting is not only a point estimate, but also includes prediction intervals to describe the range of uncertainty. In this study, a 95% prediction interval was used. Mathematically, the prediction interval is defined as:

$$\hat{y}_T + h \pm z_{\alpha/2}\hat{\sigma}_h \tag{12}$$

Where:

$\hat{y}_T$ = point forecast value for the next period h.
$z_{\alpha/2}$ = critical value of the standard normal distribution.
$\hat{\sigma}_h$ = estimated standard deviation of forecasting error for period h.

The forecasting results are visualized in the form of a time series graph that combines historical data and shaded areas to represent the prediction interval, thereby facilitating the interpretation of trends and risks of uncertainty in the future [16].

4 Results and Discussion

4.1 Description of Data and Time Series Patterns

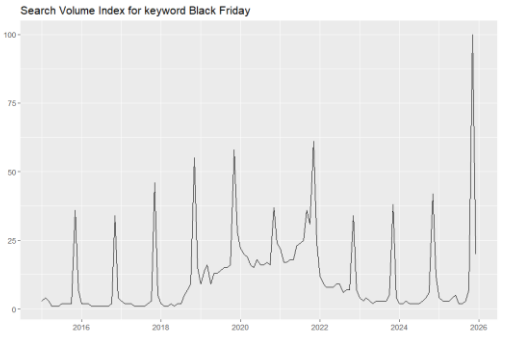


Fig. 1. Time Series Pattern

Figure 1 shows the monthly development of the Black Friday keyword Search Volume Index during the observation period. The time series is non-stationary and exhibits a very strong seasonal pattern, marked by consistent sharp spikes every November. These spikes reflect the dominant influence of Black Friday calendar variations, while in other months the index values are relatively low and stable. In addition to seasonal patterns, the data also shows a long-term trend change, with an increase until around 2020–2021, followed by a decline and an extreme spike at the end of the period. This pattern indicates that the trend is not entirely deterministic and may contain a stochastic component. Conceptually, the data structure consists of deterministic components that include trends, annual seasonality, and calendar variations, as well as stochastic components that represent random fluctuations and serial dependence. Based on these characteristics, time series regression approaches, ARIMA with exogenous variables, and hybrid models are relevant for modeling this data.

4.2 Model Comparison and Model Performance Evaluation

In this section, time series modeling is performed using several approaches, namely the regression model with ARIMA (ARIMAX) errors, the time series regression (TSR) model, and a hybrid model that combines TSR with ARMA-type errors. Each model is evaluated based on the model structure, general parameter estimation results, and prediction performance on training and testing data.

Regression Models with ARIMA Errors

M1 : Deterministic Trend & Deterministic Seasonality (Fourier)

The M1 model was constructed using regression with deterministic trends, deterministic seasonal components based on Fourier functions, and Black Friday calendar variation variables. The stochastic residual structure was modeled using ARIMA(3,0,0).

Table 2. M1 Result

Parameters	Estimate	Std. Error	Description
Trend	0.097	0.052	Positive trend
Black Friday	31.68	2.05	Significant
AR(1)	0.709	0.09	Strong autocorrelation
AR(2)	0.057	0.111	Weak autocorrelation
AR(3)	0.164	0.09	Moderate autocorrelation
S1-12	-1.197	0.683	Seasonal
C1-12	2.176	0.716	Seasonal

⋮	⋮	⋮	⋮
S5-12	-0.219	0.315	Seasonal
C5-12	0.179	0.396	Seasonal

The estimation results show that the trend has a positive effect on search volume, while the Fourier seasonal component is able to capture annual patterns flexibly. The Black Friday variable has a large positive coefficient, indicating a significant surge during that period. However, the predictive performance of this model is still moderate compared to several alternative models.

M2 : Stochastic Trend & Deterministic Seasonality (Fourier)
The M2 model uses stochastic trends through one-time differentiation with deterministic Fourier-based seasonality and Black Friday variables, as well as an ARIMA(0,1,1) error structure.

Table 3. M2 Result

Parameters	Estimate	Std. Error	Description
Black Friday	31.58	2.45	Strong calendar effect
MA(1)	−0.287	0.094	Short-term error correction
S1-12 (Fourier)	−1.224	0.702	Annual seasonal component
C1-12 (Fourier)	2.201	0.751	Annual seasonal component
⋮	⋮	⋮	⋮
S5-12 (Fourier)	−0.229	0.355	Smooth seasonal
C5-12 (Fourier)	0.173	0.457	Smooth seasonal

The estimation results show that the first-order moving average component plays a role in capturing short-term error corrections. The Black Friday variable has a large positive coefficient, indicating a significant surge in search volume during that period. Compared to Model M1, which uses deterministic trends, Model M2 is more flexible in capturing long-term level changes because trends are modeled stochastically, thus generally providing a better representation of data dynamics.

M3 : Deterministic Trend & Stochastic Seasonality
The M3 model combines deterministic trends with stochastic seasonality using ARIMA(1,0,0)(1,0,0)[12]

Table 4. M3 Result

Parameters	Estimate	Std. Error	Description
Trend	0.113	0.042	Positive trend
Black Friday	32.762	1.088	Strong calendar effect
AR(1)	0.869	0.043	Strong autocorrelation
SAR(1) [12]	0.209	0.088	Annual seasonal dependence

The estimation results show that the trend and annual seasonal components have a significant effect, and the Black Friday variable has a very strong impact. However, the prediction error of this model is relatively larger than that of Models M1 and M2, indicating that stochastic seasonality alone is not sufficient to capture the extreme seasonal patterns in the data.

M4 : Stochastic Trend & Stochastic Seasonality
The M4 model fully models trends and seasonality as stochastic components using ARIMA(0,1,1)(1,0,0)[12] with calendar variables.

Table 5. M4 Result

Parameters	Estimate	Std. Error	Description
Black Friday	33.295	1.147	Strong calendar effect
MA(1)	−0.227	0.0983	Strong autocorrelation
SAR(1) [12]	0.1804	0.088	Annual seasonal dependence

This model is capable of capturing both short-term and annual seasonal dependencies simultaneously. However, its predictive performance has not shown significant improvement compared to deterministic seasonal models, so its effectiveness remains limited.

Time Series Regression Models
M5 : TSR with Deterministic Trends
The M5 model is a simple time series regression that only includes deterministic trends.

Table 6. M5 Result

Parameters	Estimate	Std. Error	Description
Intercept	7.118	2.339	Initial level
Trend (t)	0.067	0.034	Positive trend

The estimation results show that the trend has a positive and significant effect on search volume. However, this model is unable to capture seasonal patterns and calendar variations, resulting in large residuals and poor prediction performance.

M6 : TSR with Deterministic Trends and Seasonality
Model M6 builds on Model M5 by adding a monthly dummy to capture seasonal effects.

Table 7. M6 Result

Parameters	Estimate	Std. Error	P-value	Description
Trend (t)	0.053	0.022	0.016	Significant
Month 1	5.193	2.850	0.071	Marginal
Month 2	4.440	2.859	0.123	Not significant
Month 3	4.487	2.868	0.121	Not significant
Month 4	3.134	2.878	0.279	Not significant
Month 5	3.081	2.887	0.288	Not significant
Month 6	4.128	2.897	0.157	Not significant
Month 7	4.175	2.907	0.154	Not significant
Month 8	4.423	2.917	0.132	Not significant
Month 9	6.070	2.927	0.040	Significant
Month 10	6.317	2.937	0.034	Significant
Month 11	40.664	2.947	< 0.001	Very significant
Month 12	9.511	2.957	0.002	Significant

The estimation results show that the trend remains significant and that some months, especially towards the end of the year, have a strong influence. November shows a very large coefficient, reflecting an extreme spike in search volume. The addition of seasonal components significantly improves the model's performance compared to the M5 model.

M7 : TSR with Deterministic Trends and Seasonality + Calendar Variation
Model M7 adds the Black Friday calendar variable to M6.

Table 8. M7 Result

Parameters	Estimate	Std. Error	P-value	Description
Trend (t)	0.053	0.022	0.016	Significant
Month 1	5.193	2.850	0.071	Marginal
Month 2	4.440	2.859	0.123	Not significant
Month 3	4.487	2.868	0.121	Not significant
Month 4	3.134	2.878	0.279	Not significant
Month 5	3.081	2.887	0.288	Not significant
Month 6	4.128	2.897	0.157	Not significant
Month 7	4.175	2.907	0.154	Not significant
Month 8	4.423	2.917	0.132	Not significant
Month 9	6.070	2.927	0.040	Significant
Month 10	6.317	2.937	0.034	Significant
Month 11	40.664	2.947	< 0.001	Very significant
Month 12	9.511	2.957	0.002	Significant
Black Friday (BF)	–	–	–	Not estimated

However, the calendar effect cannot be explicitly estimated due to overlap with the November dummy variable. Nevertheless, the deterministic structure of this model is able to represent the main patterns of the data well and serves as the basis for the development of a hybrid model.

Hybrid TSR Model with ARMA Errors
M8 : TSR Hybrid Model with ARMA Error
Model M8 was constructed by modeling the residuals of Model M7 using ARIMA(0,1,1). This approach aims to capture the autocorrelation that remains after the deterministic components are modeled.

Table 9. M8 Result

Parameter	Estimate	Std. Error	Description
MA(1)	-0.286	0.093	Autokorelasi residual jangka pendek

The estimation results show that the addition of the ARMA component can improve the stochastic residual structure, so that this hybrid model conceptually combines the interpretability advantages of TSR and the flexibility of ARMA.

Model Performance Comparison on Training Data
Upon estimating all models from M1 to M8, the subsequent step is to evaluate the performance of each model according to its accuracy on the training dataset. The evaluation of model performance on training data is carried out to measure how well each model captures historical data trends. The assessment metrics utilized consist of RMSE, MAPE, and sMAPE. Lower values suggest reduced prediction error rates and improved model alignment with the data.

Table 10. Model Performance Comparison on Training Data

Model	RMSE	MAPE	sMAPE
Deterministic Trend & Seasonality (M1)	3.38	0.312	0.289
Stochastic Trend & Deterministic Seasonality (M2)	3.45	0.306	0.301
Deterministic Trend & Stochastic Seasonality (M3)	3.70	0.338	0.286
Stochastic Trend & Seasonality (M4)	3.74	0.308	0.275
Time Series Regression with Deterministic Trends (M5)	12.60	2.51	0.929
TSR with Deterministic Trends and Seasonality (M6)	7.73	1.55	0.786
TSR with Deterministic Trends and Seasonality + Calendar Variation (M7)	7.73	1.55	0.786
TSR Hybrid Model with ARMA Error (M8)	3.45	0.308	0.296

The evaluation results show that the TSR models (M6 and M7) provide a significant improvement in performance compared to the simple TSR model (M5), especially after seasonal components and calendar variations are included. This confirms that seasonal patterns and the Black Friday effect are the main determinants in the data. ARIMAX models with deterministic seasonality (M1 and M2) are able to capture annual patterns flexibly, but the complexity of the model is relatively high. Models with pure stochastic seasonality (M3 and M4) show less than optimal performance in dealing with extreme spikes in November. Model M8 (Hybrid TSR with ARMA Error) showed the best overall performance. This model not only maintains the interpretability of deterministic components but also improves the residual structure through stochastic modeling, resulting in the smallest prediction error and residuals that are close to white noise.

To strengthen the quantitative evaluation, a visualization comparing the actual data and the predicted results from all models M1 to M8 during the training period was performed.

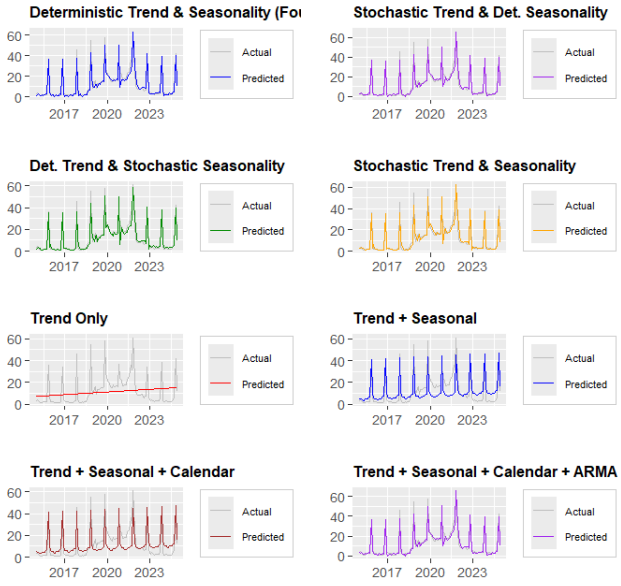


Fig. 2. Comparison of Actual Data Train and Model Predictions

Based on the training data plot, it can be seen that models with seasonal components are able to follow actual patterns much better than models without seasonal components. The Deterministic Trend & Seasonality (M1) and Stochastic Trend & Deterministic

Seasonality (M2) models show good fit in capturing the pattern of recurring annual spikes, although there is still a slight underestimation in some peak periods. Models with stochastic seasonality (M3 and M4) are also able to follow the general pattern of the data, but their predictions appear smoother and less sharp at extreme peaks. This indicates that the stochastic seasonal approach alone is not yet fully optimal for representing very strong seasonal spikes. Visually, the four ARIMAX models have successfully captured the main structure of the training data. This shows that the combination of trend, seasonality, and stochastic components is relevant for this data.

In contrast, the Trend Only (M5) model clearly cannot represent the training data because it fails to capture annual seasonal patterns and extreme spikes. The TSR model with trends and seasonality (M6) shows a significant improvement, where seasonal patterns already follow the actual data, although there are still deviations at some peaks. The addition of calendar variation in M7 makes the predictions closer to the actual patterns, especially in periods with high spikes. The TSR + ARMA hybrid model (M8) shows smoother predictions and is more responsive to short-term fluctuations than pure TSR. Visually, M8 appears to be able to balance seasonal pattern fit and residual stability. However, a final assessment cannot be made based solely on the training plot and needs to be confirmed through testing data evaluation and white noise residual analysis.

Model Performance Comparison on Testing Data

Table 11. Model Performance Comparison on Testing Data

Model	RMSE	MAPE	sMAPE
Deterministic Trend & Seasonality (M1)	16.3	1.35	0.745
Stochastic Trend & Deterministic Seasonality (M2)	17.0	0.916	0.606
Deterministic Trend & Stochastic Seasonality (M3)	17.4	2.44	1.01
Stochastic Trend & Seasonality (M4)	17.3	2.11	0.96
Time Series Regression with Deterministic Trends (M5)	26.7	3.36	1.19
TSR with Deterministic Trends and Seasonality (M6)	16.7	2.16	0.950
TSR with Deterministic Trends and Seasonality + Calendar Variation (M7)	16.7	2.16	0.950
TSR Hybrid Model with ARMA Error (M8)	16.8	1.03	0.652

The evaluation results show that the TSR models (M6 and M7) provide a significant improvement in performance compared to the simple TSR model (M5), especially after seasonal components and calendar variations are included. This confirms that seasonal patterns and the Black Friday effect are the main determinants in the data. ARIMAX models with deterministic seasonality (M1 and M2) are able to capture annual patterns flexibly, but the complexity of the model is relatively high. Models with pure stochastic seasonality (M3 and M4) show less than optimal performance in dealing with extreme spikes in November. Model M8 (Hybrid TSR with ARMA Error) showed the best overall performance. This model not only maintains the interpretability of deterministic components but also improves the residual structure through stochastic modeling, resulting in the smallest prediction error and residuals that are close to white noise.

To strengthen the quantitative evaluation, a visualization comparing the actual data and the predicted results from all models M1 to M8 during the training period was performed.

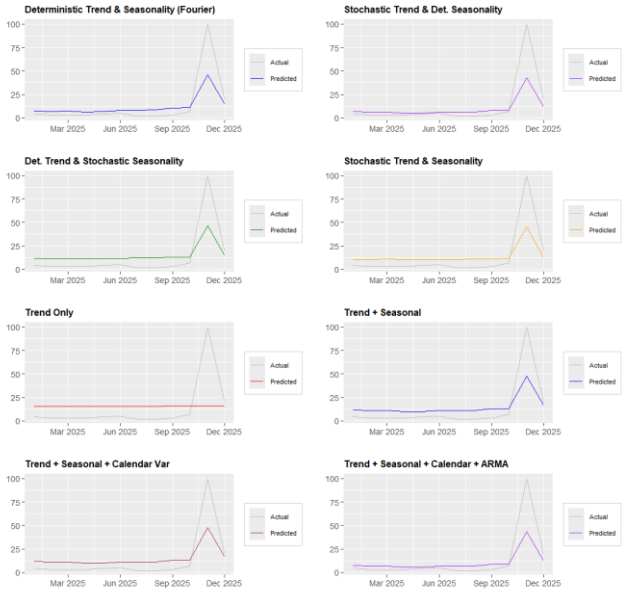


Fig. 3. Comparison of Actual Data Test and Model Predictions

Visually, the testing data shows an extreme spike at the end of the period, which poses a major challenge for all models. The Trend model alone (M5) clearly fails to follow the actual pattern because its predictions are almost flat and very far from the actual values. The TSR model with trends and seasonality (M6–M7) is able to capture the upward trend towards the end of the period, but still tends to underestimate the magnitude of the spike. The ARIMAX model with seasonal components (M1–M4) showed prediction patterns that were more in line with the dynamics of the actual data, although some models still produced peaks that were too low or too smooth. This shows that the seasonal structure helps in recognizing the timing of the spike, but does not fully capture its intensity. In general, models with seasonal components clearly outperformed models without seasonal components in the test data.

The hybrid TSR model with ARMA error (M8) shows relatively more adaptive predictions than pure TSR, especially in following pattern changes ahead of peaks. The addition of ARMA components to the residuals helps improve the model's response to short-term dynamics in out-of-sample data. However, differences in magnitude between predictions and actual data are still apparent, especially at extreme peaks. This indicates that seasonal events with very high intensity remain difficult to predict with precision. Therefore, although some models appear more stable visually on the testing data, this interpretation is not sufficient to determine the best model. Evaluation of error metrics and white noise residual tests are still necessary as a basis for final decision making.

4.3 Residual Diagnostic Test

In this section, a residual diagnostic test is performed to assess whether each model has been able to capture all dependency patterns in the data. The test is performed using the Ljung–Box test with the null hypothesis that the residuals are white noise (there is no autocorrelation up to a certain lag). A good model is expected to produce residuals that are not correlated with each other.

Table 12. Model Performance Comparison on Testing Data

Model	Model Specifications	Ljung-Box Statistics	P-value	Residual White Noise
M1	Deterministic Trend & Deterministic Seasonality	24.2	0.0040	No
M2	Stochastic Trend & Deterministic Seasonality	34.9	0.0005	No
M3	Deterministic Trend & Stochastic Seasonality	24.4	0.2260	Yes
M4	Stochastic Trend & Stochastic Seasonality	23.9	0.3000	Yes
M5	Trend	168.00	0.0000	No
M6	Trend + Seasonal	718.00	0.0000	No
M7	Trend + Seasonal + Calendar	718.00	0.0000	No
M8	Trend + Seasonal + Calendar + ARMA	34.76	0.0549	Yes (marginal)

Based on the diagnostic test results, Models M1 and M2 show p-values smaller than 0.05, so the null hypothesis is rejected. This indicates that the residuals of both models still contain significant autocorrelation. Thus, although M1 and M2 are able to capture the deterministic component well, their stochastic structure is not yet fully adequate. Models M3 and M4 produced p-values greater than 0.05, so the residuals can be considered white noise. This shows that the stochastic seasonal approach in both models is quite effective in eliminating serial dependence in the residuals. Diagnostically, M3 and M4 satisfy the basic assumptions of time series models. In contrast, all pure TSR models (M5, M6, and M7) show very large Ljung–Box statistics with p-values close to zero. This indicates that the residuals still contain very strong autocorrelation. In other words, TSR models without a stochastic error component are not yet able to adequately capture the temporal dynamics of the data.

For Model M8 (TSR hybrid with ARMA error), the Ljung–Box test results on ARIMA residuals show a p-value of 0.05492, which is slightly greater than the 5% significance level. These results indicate that the M8 model residuals can be considered close to white noise. Thus, the addition of the ARMA component to the TSR residuals has been proven to improve the stochastic structure of the model. Overall, the diagnostic test shows that stochastic error modeling is an important component in this data. Models that do not include the ARMA structure tend to produce correlated residuals, while models with ARIMA errors or hybrid approaches provide better diagnostic results and are worth considering at the stage of selecting the best model.

4.4 Best Model Selection

To summarize the results of performance evaluation and residual diagnostic tests, a final comparison of all tested models was conducted. This summary was used as the basis for determining the best model to be used in the forecasting stage. The comparison results are presented in Table 13.

Table 13. Best Model Selection

Model	RMSE Test	MAPE Test	White Noise Residual	Description
M1	16.3	1.35	No	Strong seasonal factors, residual problems
M2	17.0	0.92	No	More flexible, but still autocorrelated
M3	17.4	2.44	Yes	Low accuracy
M4	17.3	2.11	Yes	Parsimonious, limited accuracy
M5	26.7	3.36	No	The model is too simplistic
M6	16.7	2.16	No	Deterministic only
M7	16.7	2.16	No	Residual still correlates
M8	16.8	1.03	Yes	Selected model

Based on the evaluation results of the test data and residual diagnostic tests, the best model was selected by considering two main aspects, namely the generalization ability of the model and the adequacy of the white noise assumption in the residuals. Evaluation of the test data shows that models with simpler trend and seasonal structures tend to produce more stable prediction errors than models without seasonal or calendar components. The trend-only model (M5) had much higher RMSE and MAPE values, making it less capable of capturing the seasonal patterns in the data. Meanwhile, models with the addition of seasonal and calendar variation components show improved prediction accuracy, although not always significant for all evaluation metrics. This indicates that the existence of seasonal patterns plays an important role in explaining data dynamics. Therefore, models with a combination of trends and seasonal patterns are the main candidates for further consideration.

From a residual diagnostic perspective, the Ljung–Box test results indicate that not all models satisfy the white noise residual assumption. Models M3 and M4 have p-values above the 5% significance level, so their residuals can be considered to have no significant autocorrelation. For model M8 (Trend + Seasonal + Calendar + ARMA), the Ljung–Box test results in a p-value of 0.0549, which is slightly above the 5% threshold, so the residuals can be considered to be close to white noise. This indicates that the addition of the ARMA component to the residuals successfully reduces the remaining dependency patterns. Considering the balance between test data accuracy and residual assumption validity, model M8 is considered the most appropriate to use as the final model.

4.5 Forecasting with The Best Model

Based on the performance evaluation results on the test data and residual diagnostic tests in Section 4.4, the M8 model (TSR with Trend, Seasonality, Calendar Variation, and ARMA Error) was selected as the best model for forecasting. This model has relatively stable predictive performance on the test data and residuals that are close to white noise based on the Ljung–Box test. Thus, the M8 model is considered capable of representing the deterministic and stochastic components in the data more evenly. The forecasting process is carried out using model parameters estimated from all historical data. Forecasting focuses on the period after the last observation data. The forecasting results are presented in graph form to facilitate interpretation of future patterns.

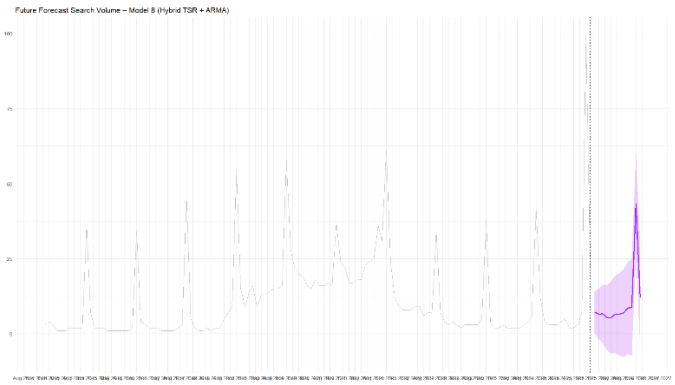


Fig. 4. Forecasting with The Best Model

The forecasting results show that the predicted values follow a seasonal pattern consistent with historical data, particularly recurring spikes at the end of the year. This indicates that the seasonal components and calendar variations incorporated into the model continue to play an important role in determining the forecast values. In addition, the predicted levels are within a reasonable range compared to historical fluctuations, without showing explosive behavior or unrealistic trends. However, forecasting uncertainty increases as the time horizon lengthens, as reflected in the variation in forecast values. Therefore, these forecasting results should be used as an indication of general direction and patterns, rather than as definitive values. Overall, the M8 model provides forecasting results that are consistent with the characteristics of the data and are suitable for prospective analysis.

5 Conclusions and Recommendations

5.1 Conclusion

Based on the results of the analysis in Chapter 4, it can be concluded that the Search Volume Index data for the keyword Black Friday has complex time series characteristics, marked by long-term trends, very strong annual seasonal patterns, and the dominant influence of calendar variations in November. A comparison of various model specifications shows that models with simple structures, such as TSR without seasonal or calendar components, are not yet able to adequately capture the dynamics of the data and produce residuals that still contain autocorrelation. Regression models with ARIMA errors and hybrid models provide improved performance because they are able to combine deterministic and stochastic components. Based on performance evaluation on training and testing data and residual diagnostic test results, the TSR model with trend, seasonality, calendar variation, and ARMA error (M8) shows the most stable performance and residuals that are close to white noise. Therefore, the M8 model is selected as the best model and is used to forecast the upcoming period.

5.2 Suggestions

For further research development, it is recommended to use data with higher frequency so that seasonal patterns and calendar effects can be captured in greater detail. In addition, the addition of other relevant exogenous variables, such as online promotions or macroeconomic factors, can be considered to improve the accuracy of the model. Comparisons with other approaches, such as nonlinear models or machine learning methods, can also be made as a comparison to test the robustness of the forecasting results.

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